Memory Leak Detection Tool

A Robust Solution for Runtime Heap Monitoring

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EXECUTIVE SUMMARY

Professional Library

We built a production-ready system library for Linux that identifies memory leaks with bit-perfect accuracy.

Complete Monitoring

Our solution tracks every heap allocation and deallocation to ensure no memory is ever lost or orphaned.

Actionable Data

The library provides developers with clear diagnostic metadata, helping find and fix code errors instantly.

PROCESS MEMORY ARCHITECTURE

Stack Segment

Local variables and function frames. Automatically managed by the OS. Fast, fixed-size, and highly predictable.

Heap Segment

Dynamic memory requested via malloc(). Decided manually by the developer. This is the primary region for memory leaks.

Collision Risk

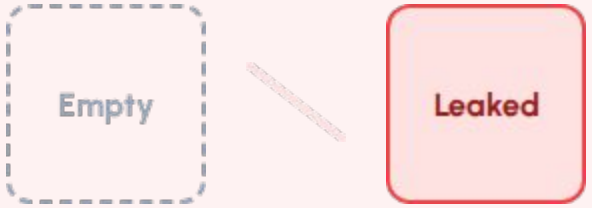
Stack (grows down) and Heap (grows up) share the same space. Collision leads to catastrophic system failure and crashes.

ANATOMY OF A MEMORY LEAK

State 1: Active

Pointer on the stack correctly references the block on the heap.

State 2: Leaked



Pointer is lost, but heap memory remains unreachable and permanently occupied.

THE BUSINESS IMPACT

Web Server

2.8 GB

Loss after 30 days based on a tiny 100-byte leak per incoming request.

IoT Device

Weeks

Timeline to total crash for 64MB embedded sensors from small reading leaks.

Background Daemon

700 MB

Consumed after 500k background queries, causing severe system slowdown.

CORE DETECTION ENGINE

Preprocessor Macros

Intercepts malloc at compile-time to inject precise source code location metadata (__FILE__, __LINE__).

Dynamic Hooking

Uses dlsym to route calls to the actual libc while recording metadata silently in the background.

Thread Safety

Mutex-protected global linked list ensures tracking integrity during concurrent multi-threaded allocations.

THE ALLOCATION PIPELINE

1

Trigger

Application initiates a standard malloc(size) call.

2

Metadata

Macro expands the call to include file and line number information.

3

Real Alloc

Tool calls libc malloc via dynamic linker bridge (dlsym).

4

Tracking

Mutex-protected entry created in the AllocationRecord list.

5

Result

Original pointer returned to the application transparently.

MEMORY TRACKING: MALLOC() LIFECYCLE

From Application Call to Tracked Pointer

1 TRIGGER

App calls malloc(size).

```
int *arr =
malloc(256);
```

Application

The application requests memory using malloc().

2 METADATA

Macro expands to include source location.

```
#define malloc(size)\
 tracked_malloc((size),
 __FILE__, __LINE__, __func__)
```

```
tracked_malloc(256,
 "main.c", 42, "main")
```

The macro automatically adds file, line, and function information.

3 REAL ALLOCATION

Tool calls libc malloc via dynamic linker bridge.

memtrack.so
(tracking layer)

dlsym(RTLD_NEXT, "malloc")

libc.so.6
(real malloc)

Heap Memory
Allocated

The real malloc() is resolved and called safely.

4 TRACKING

Entry created in the AllocationRecord list.

head

AllocationRecord

- id: 1
- address: 0x55f3c2a01260
- size: 256 bytes
- file: "main.c"
- line: 42
- func: "main"
- timestamp: 1713360601
- next: NULL

A record is stored in the thread-safe linked list.

5 RESULT

Original pointer returned to application.

```
int *arr = 0x55f3c2a01260;
```

Application Receives Tracked Pointer

The program gets back the valid memory pointer and continues execution.

KEY TAKEAWAY: Memory tracking happens transparently. The application code doesn't change, but every allocation is recorded with precise source information.

THE ALLOCATIONRECORD STRUCT

```
ypedef struct AllocationRecord {
    unsigned long id;
    void *address;
    size_t size;
    char file[256];
    int line;
    char func[64];
    time_t timestamp;
    struct AllocationRecord *next;
 } AllocationRecord;
```

Links metadata directly to heap pointers for real-time diagnostics.

VALIDATION SUITE: VALGRIND BENCHMARK

Test Case	Scenario	Our Tool	Valgrind Baseline	Status
test1	No Leak	0 blocks (0B)	0 blocks (0B)	PASS
test2	Single Leak	1 block (50B)	1 block (50B)	PASS
test3	Multi Leak	10 blocks (640B)	10 blocks (640B)	PASS
test4	Large Allocation	1 block (2.00MB)	1 block (2.09MB)	PASS
test5	Double Free	1 block + WARN	1 block (64B)	PASS
test6	Thread Safety	4 blocks (1 KB)	4 blocks (1 KB)	PASS

✓ 100% BIT-PERFECT ACCURACY VS VALGRIND

SYSTEM-WIDE INTEGRATION

1. System Install

```
sudo cp ./include/memtrack.h /use/local/include
sudo cp ./build/libmemtrack.so /usr/local/lib
sudo ldconfig
```

2. Easy Includes

Include memtrack.h in any project to access high-fidelity file/line metadata automatically.

3. Simple Linking

Build your projects using the -lmemtrack flag for a seamless native development experience.

Questions?

Thank you for your attention.

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